EXPERIMENT 9

SIMULATION OF CONVOLUTIONAL CODES AND ANALYSIS OF ENCODED OUTPUT BIT SEQUENCES USING MATLAB

OBJECTIVE:

(i) To implement convolutional encoding for a given binary input sequence using a rate convolutional encoder and tabulate the corresponding encoded output pairs.

(ii) To visualize and compare the input bit sequence and encoded output bit sequence using MATLAB plots to analyze the redundancy introduced by convolutional coding.

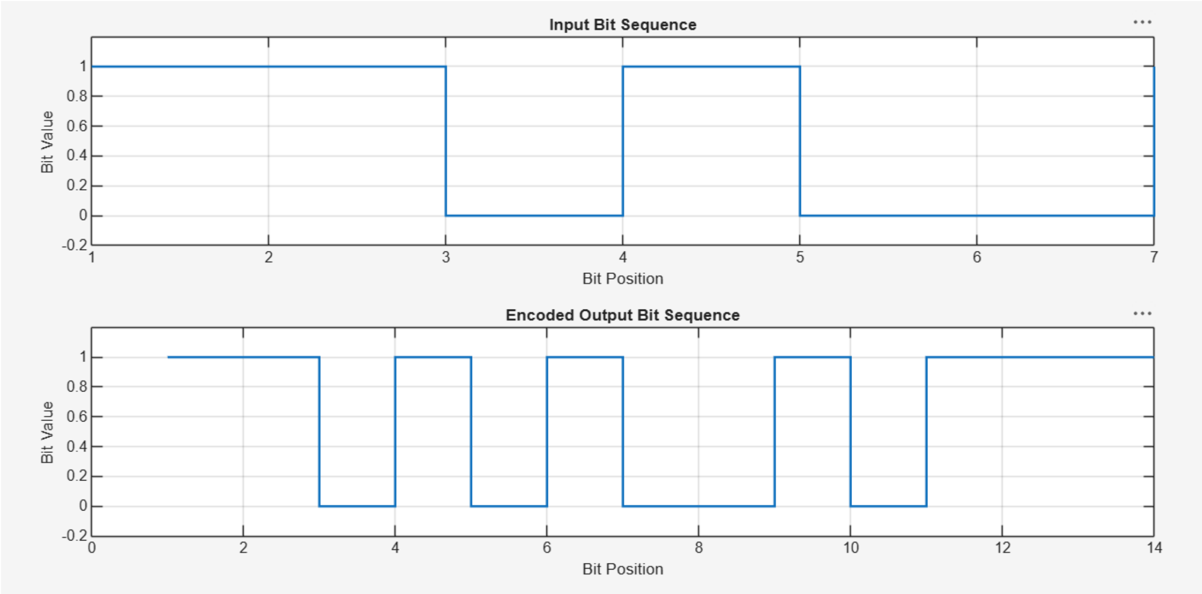
PROGRAM:

```
clc;

clear;

close all;

input_bits = [1 1 0 1 0 0 1];

trellis = poly2trellis(3,[7 5]);

encoded_bits = convenc(input_bits,trellis);

encoded_pairs = reshape(encoded_bits,2,[])';

output_pair = cell(length(input_bits),1);


for i = 1:length(input_bits)

   output_pair{i} = sprintf('%d%d', ...

       encoded_pairs(i,1), encoded_pairs(i,2));

end

% Create table

Bit_Position = (1:length(input_bits))';

Input_Bit = input_bits';

Output_Encoded = output_pair;      % <-- Do NOT transpose

T = table(Bit_Position, Input_Bit, Output_Encoded);

disp('Convolutional Encoding Table');
```

```
disp(T);

% Plot

figure;

subplot(2,1,1)

stairs(input_bits,'LineWidth',1.5)

ylim([-0.2 1.2])

title('Input Bit Sequence')

xlabel('Bit Position')

ylabel('Bit Value')

grid on

subplot(2,1,2)

stairs(encoded_bits,'LineWidth',1.5)

ylim([-0.2 1.2])

title('Encoded Output Bit Sequence')

xlabel('Bit Position')

ylabel('Bit Value')

grid on
```

OUTPUT:

```
Command Window
Convolutional Encoding Table
    Bit_Position    Input_Bit    Output_Encoded
    ____________    _________    ______________

         1              1           {'11'}
         2              1           {'01'}
         3              0           {'01'}
         4              1           {'00'}
         5              0           {'10'}
         6              0           {'11'}
         7              1           {'11'}

>>
```


Input Bit Sequence
Bit Value
Bit Position
Encoded Output Bit Sequence
Bit Value
Bit Position